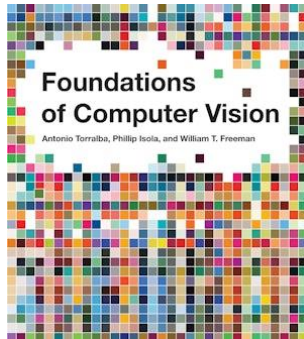


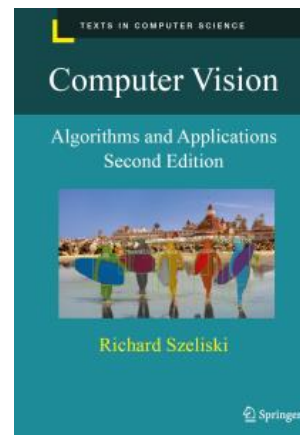
Image Formation

Dr. Praisan Padungweang, Collage of Computing, KKU

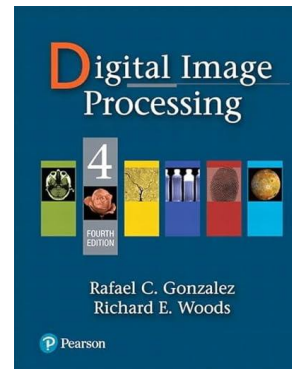
Textbook



Foundations of Computer Vision
by Torralba, Isola, Freeman

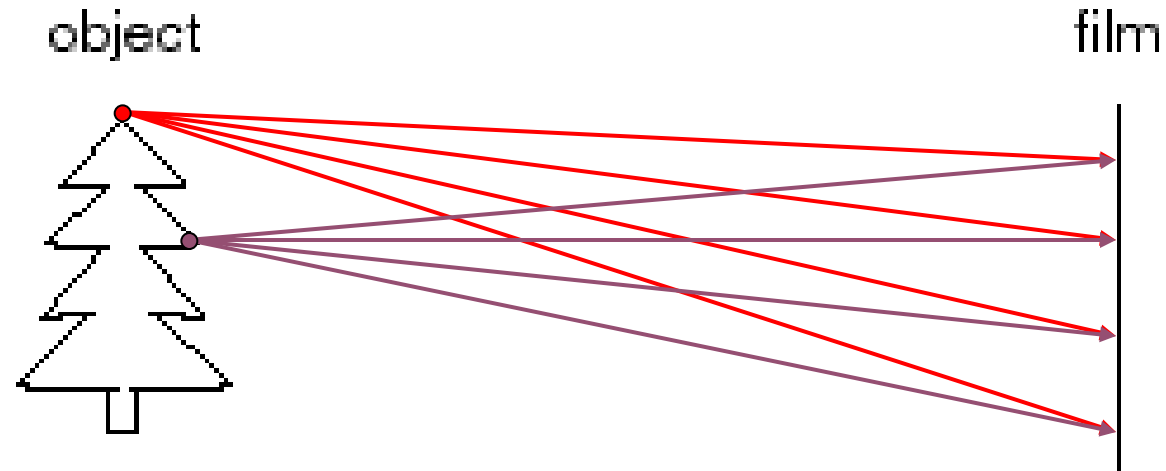


Computer Vision: Algorithms and Applications, 2nd ed.
by Richard Szeliski, <http://szeliski.org/Book/>



Digital Image Processing 4th Edition
by Rafael Gonzalez, Richard Woods

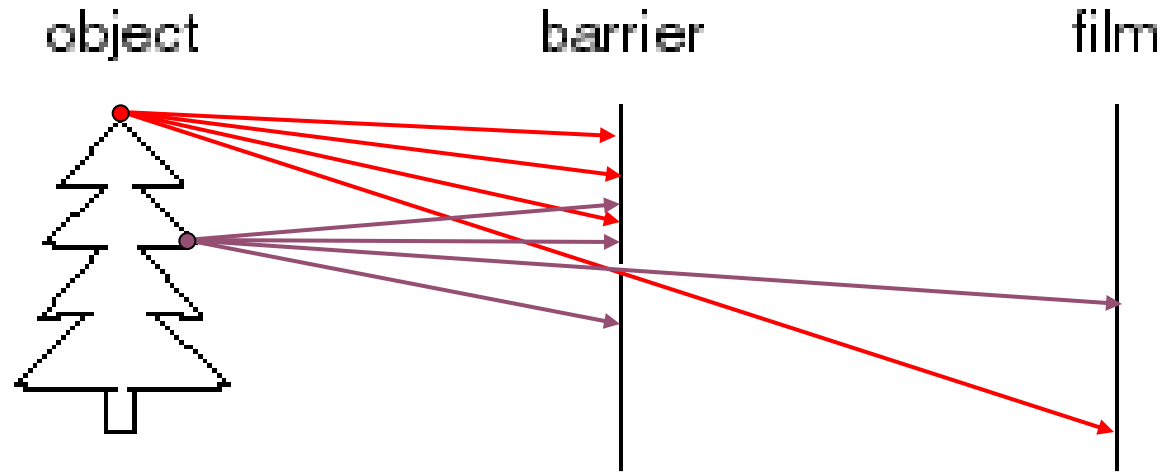
Image formation



Let's design a camera

- Idea 1: put a piece of film in front of an object
- Do we get a reasonable image?

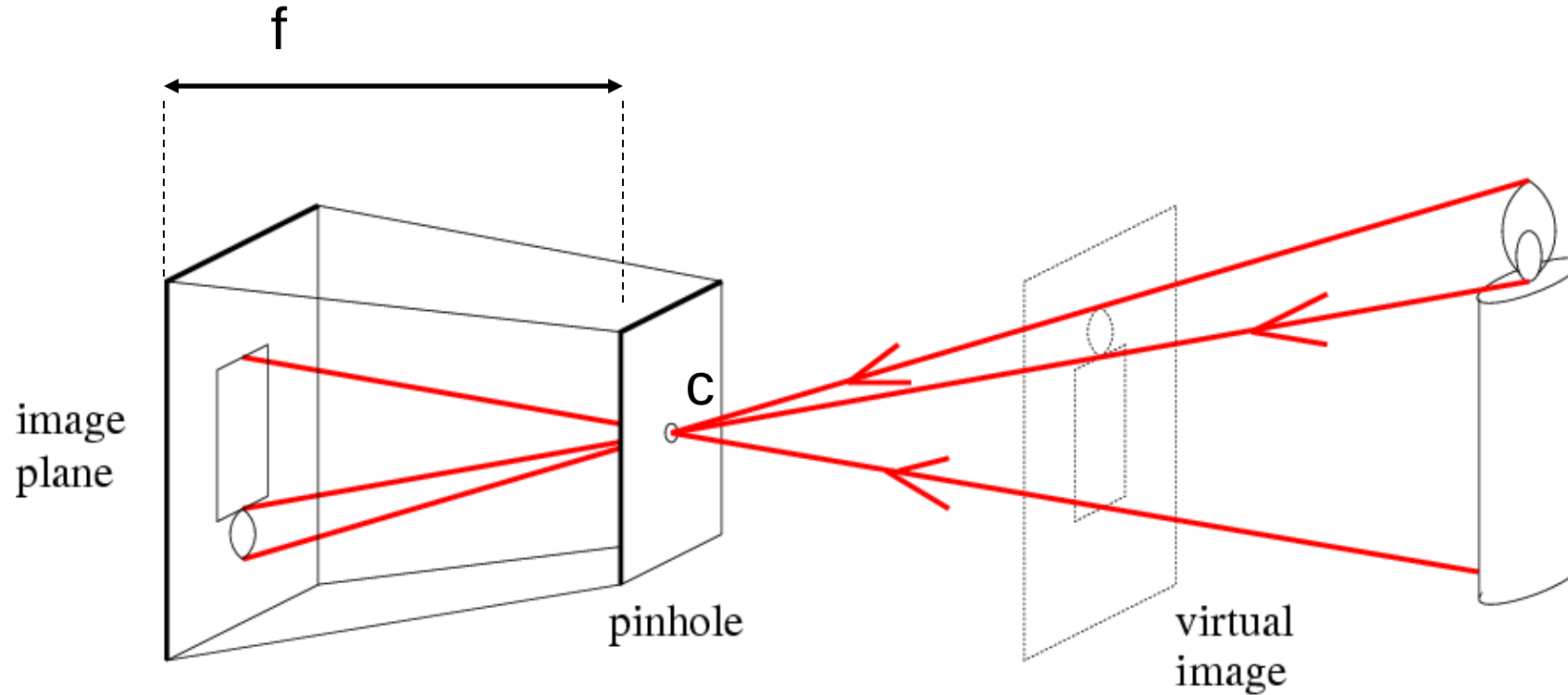
Pinhole camera



Idea 2: add a barrier to block off most of the rays

- This reduces blurring
- The opening known as the **aperture**

Pinhole camera



f = focal length
 c = center of the camera

Camera obscura: the pre-camera

- Known during classical period in China and Greece (e.g. Mo-Ti, China, 470BC to 390BC)

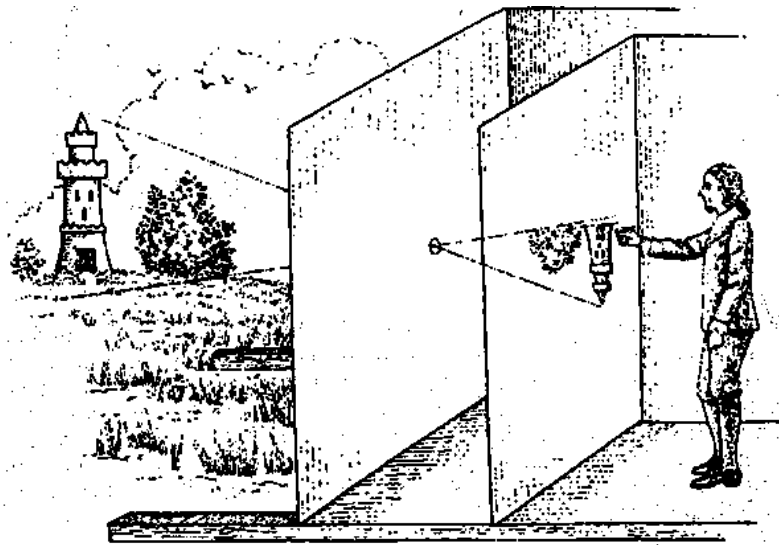


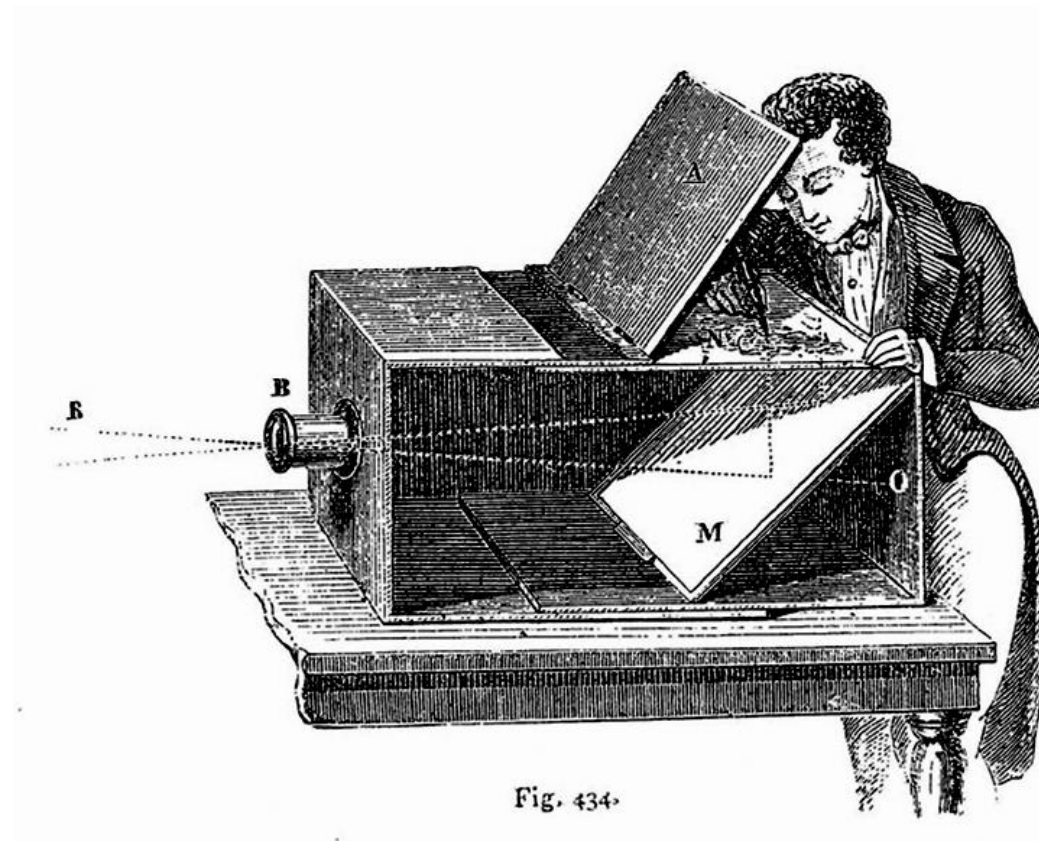
Illustration of Camera Obscura



Freestanding camera obscura at UNC Chapel Hill

Photo by Seth Ilys

Camera Obscura used for Tracing



Lens Based Camera Obscura, 1568

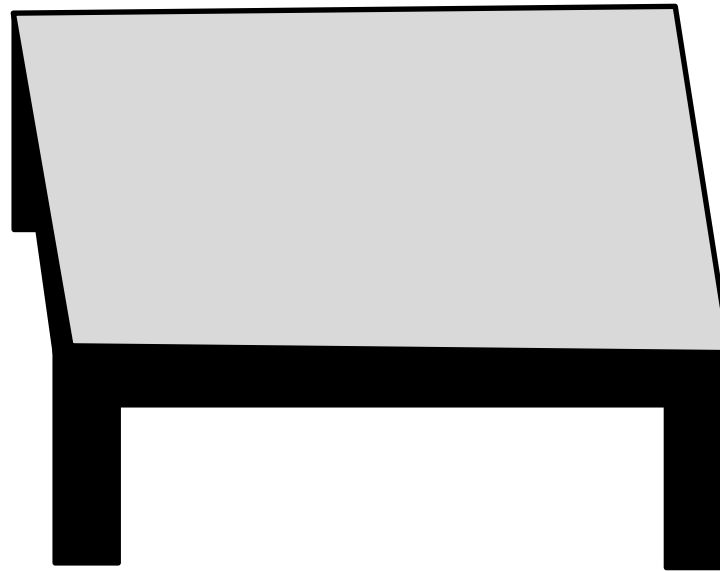
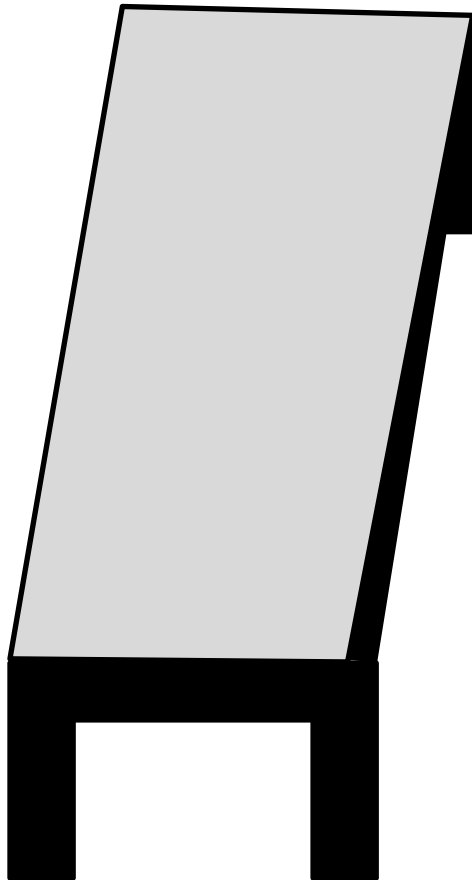
Projection can be tricky...



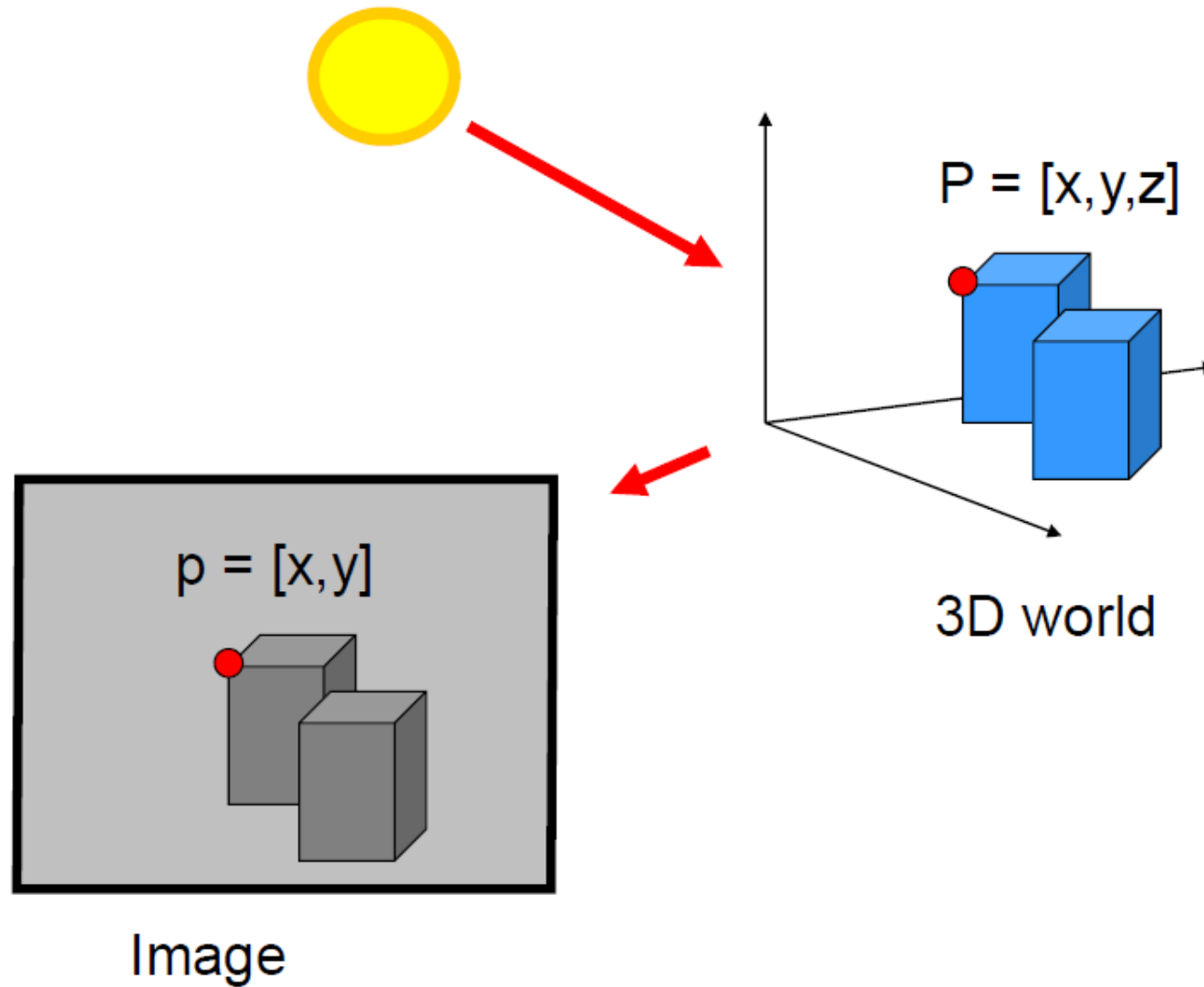
Projection can be tricky...



- https://www.youtube.com/watch?v=_EqGqEpOV3o



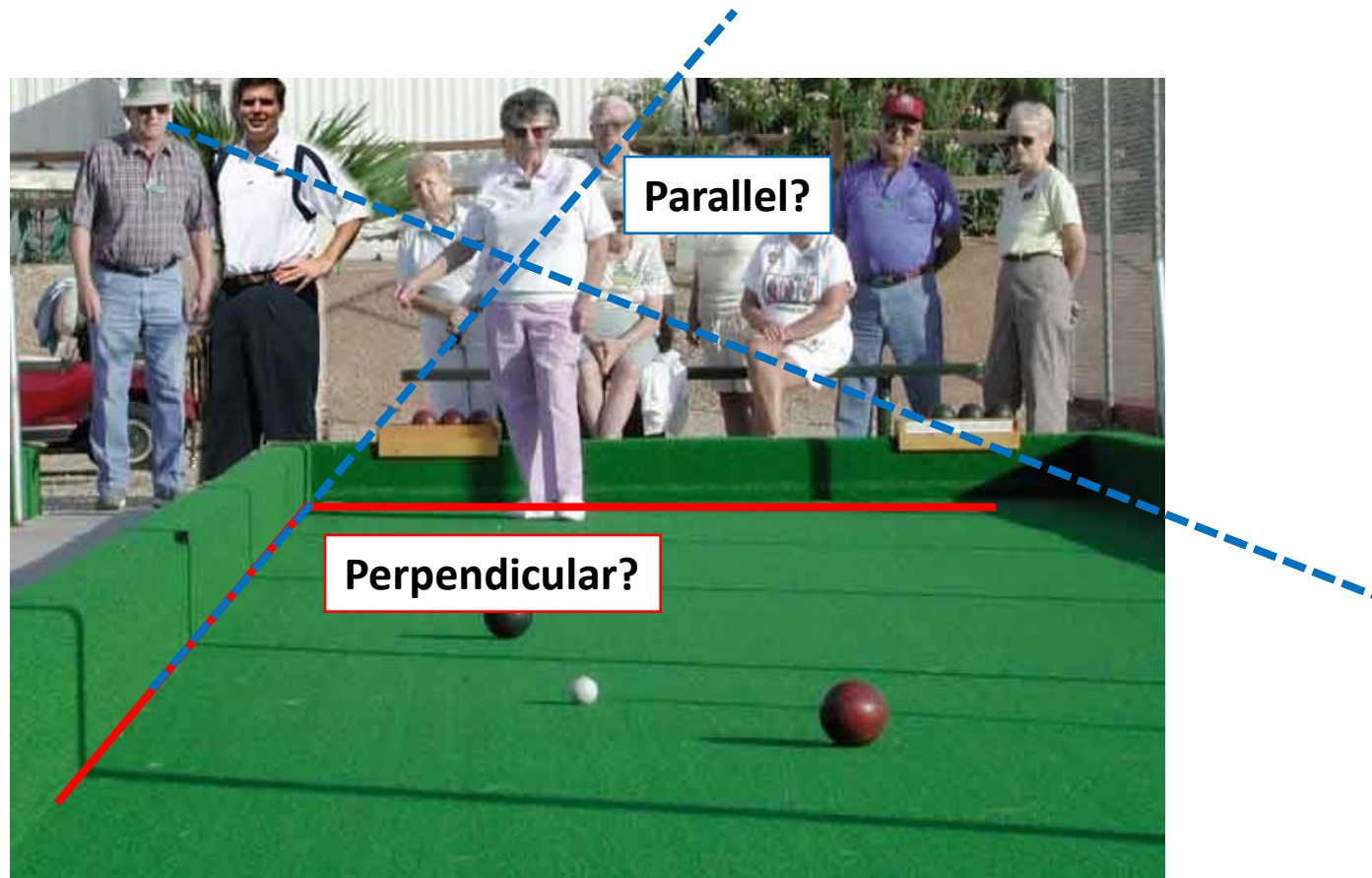
From the 3D to 2D



Projective Geometry

What is lost?

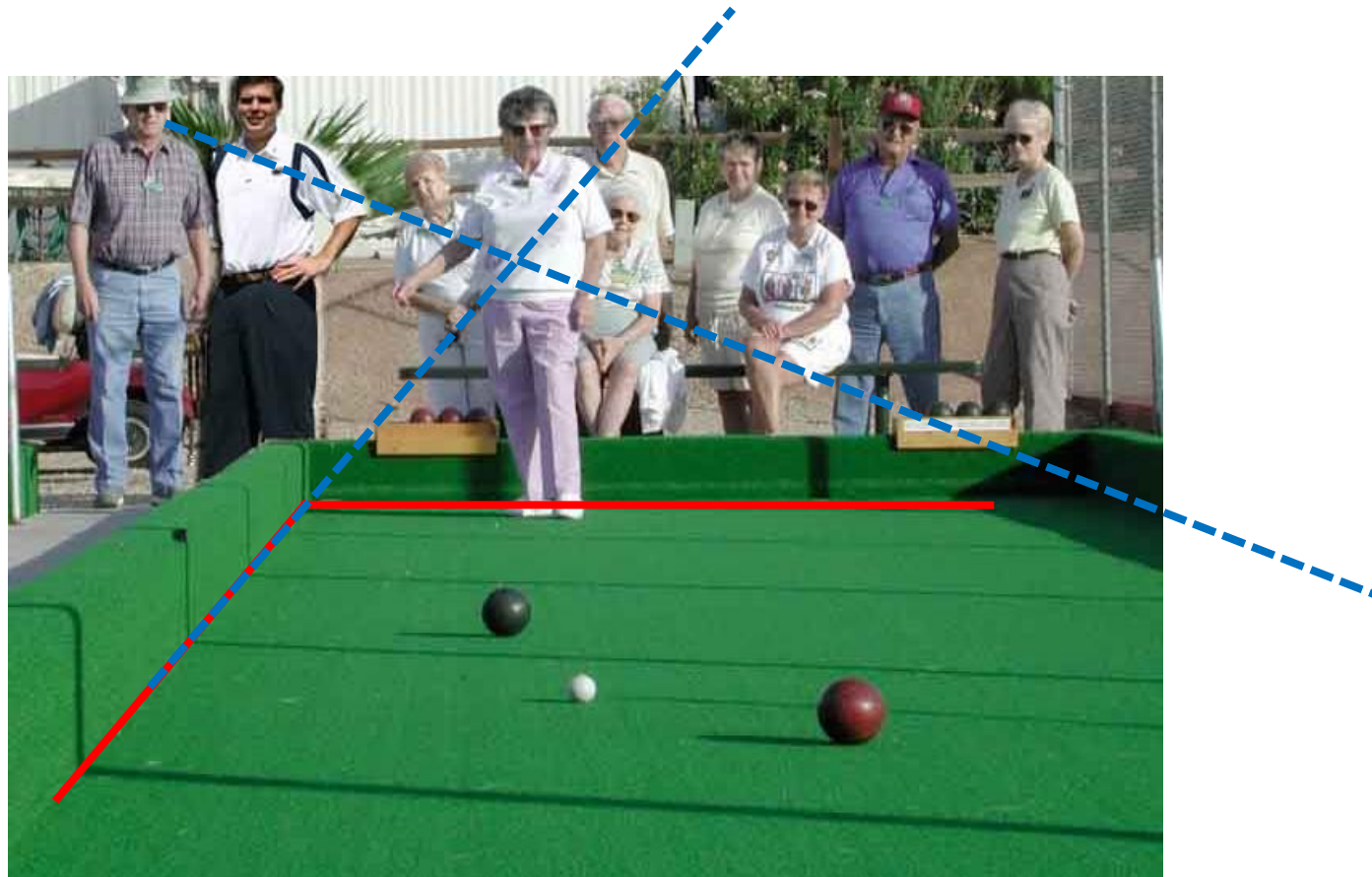
- Length
- Angles



Projective Geometry

What is preserved?

- Straight lines are still straight



Length and area are not preserved

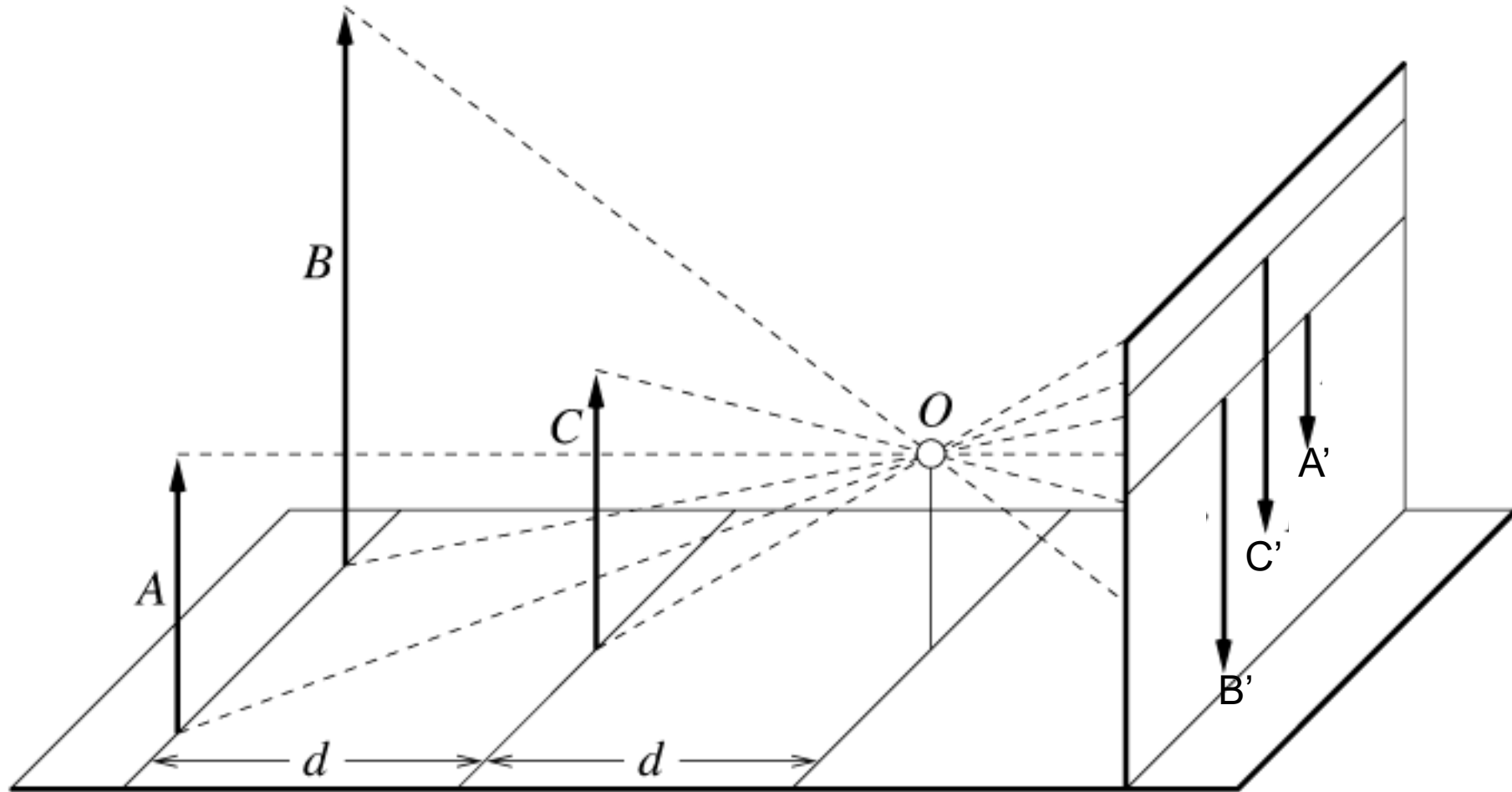
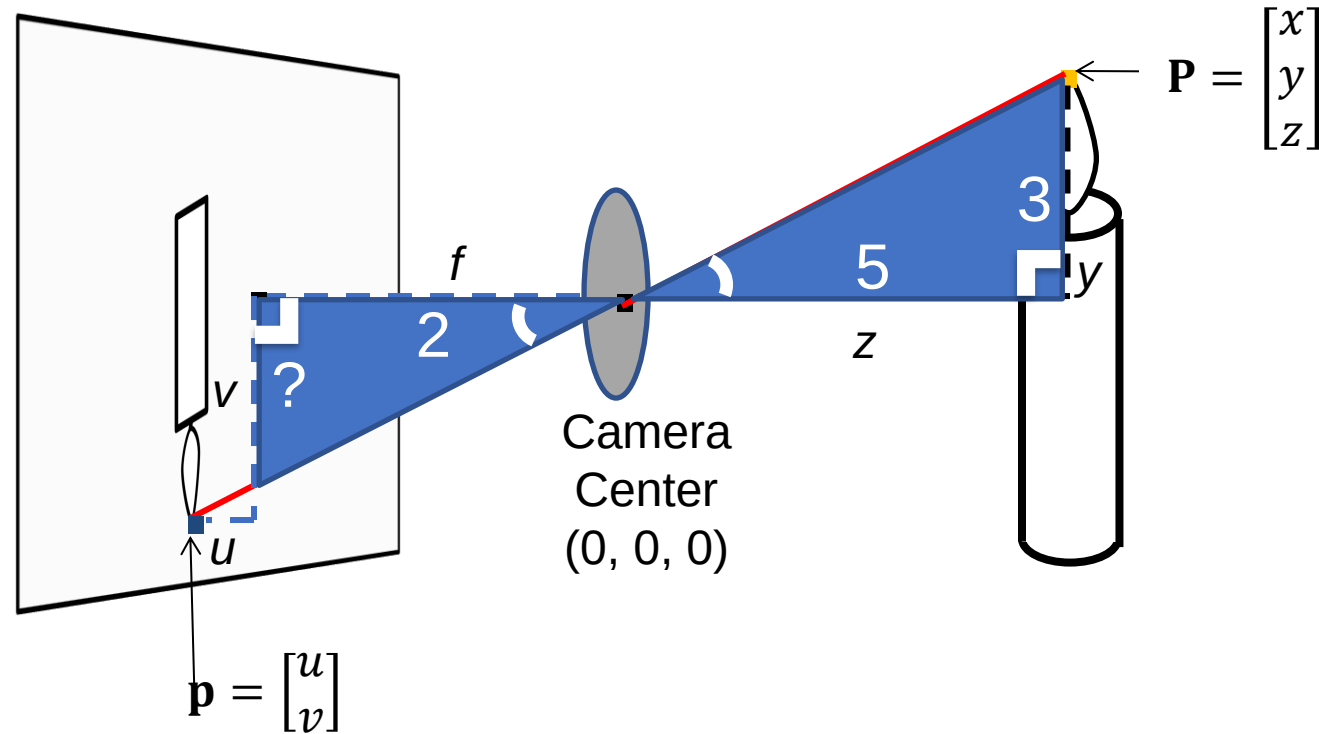


Figure by David Forsyth

Projection: world coordinates $\rightarrow$ image coordinates



If $x = 2$, $y = 3$, $z = 5$, and $f = 2$

What are u' and v' ?

$$\frac{v'}{-f} = \frac{y}{z}$$

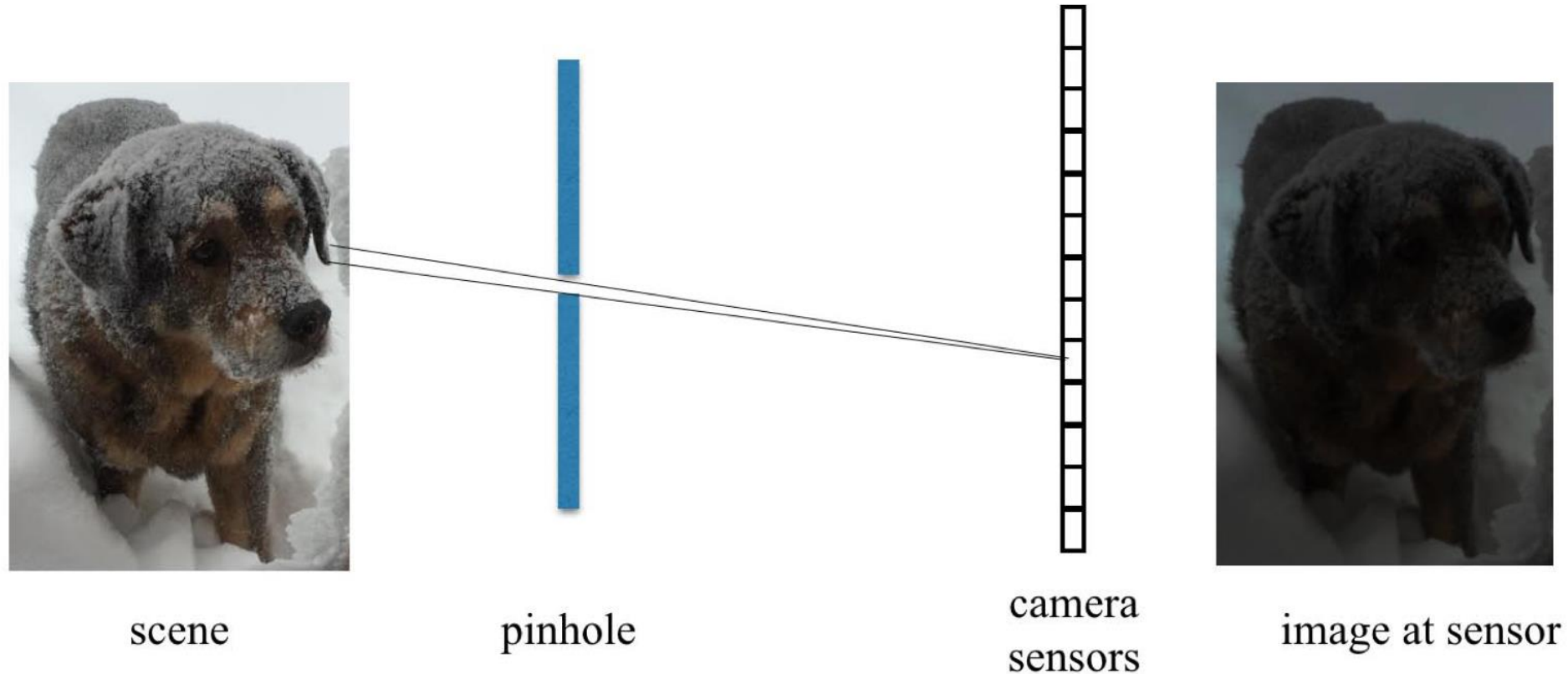
$$u' = -x * \frac{f}{z}$$

$$v' = -y * \frac{f}{z}$$

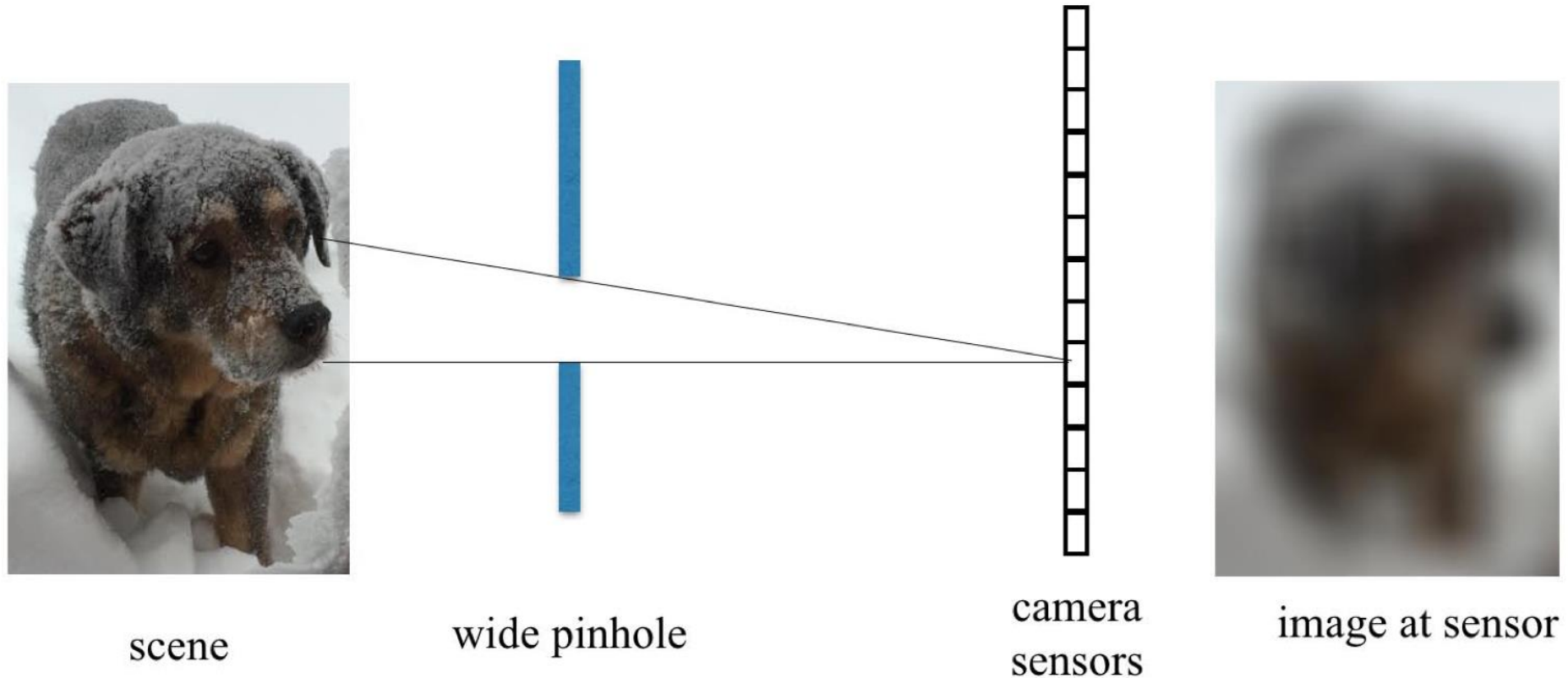
$$u' = -2 * \frac{2}{5}$$

$$v' = -3 * \frac{2}{5}$$

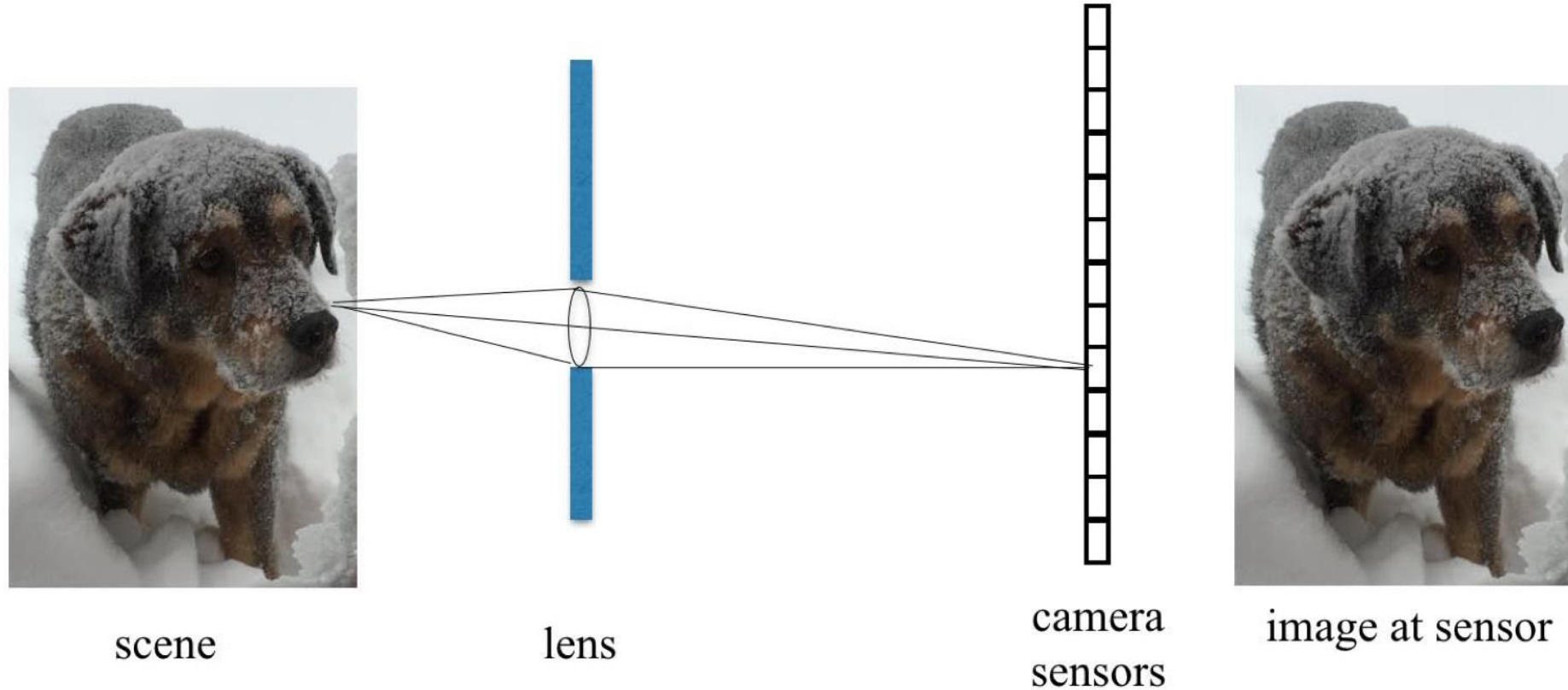
A problem: pinhole camera images are dark, or require long exposures



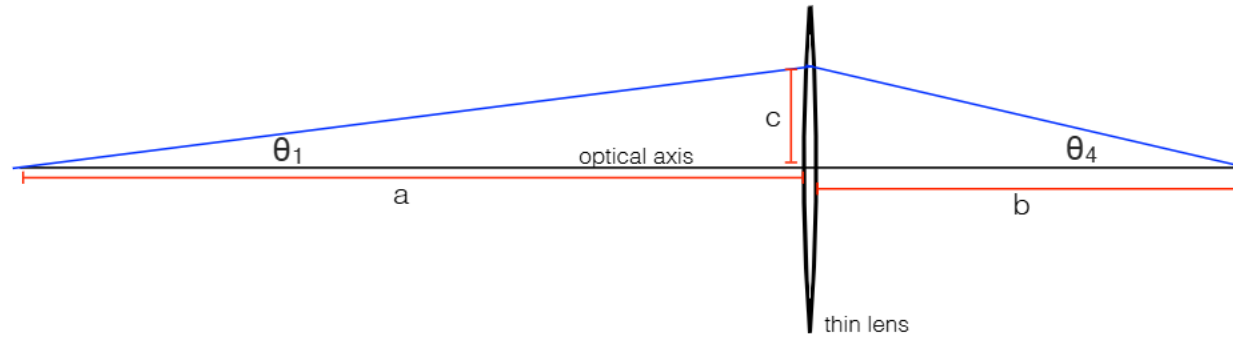
Large aperture gives a brighter image, but at the price of sharpness



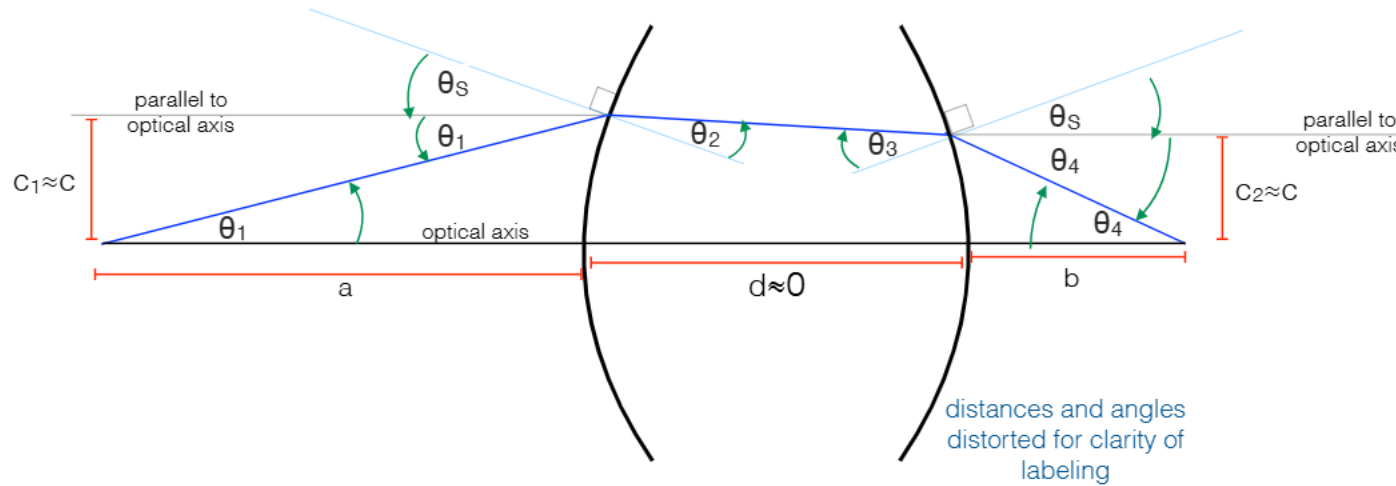
A lens allows a large aperture and a sharp image



Lens

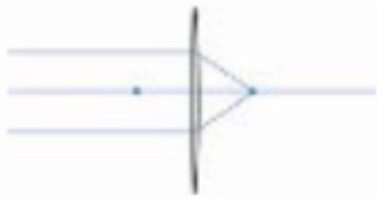


(a)

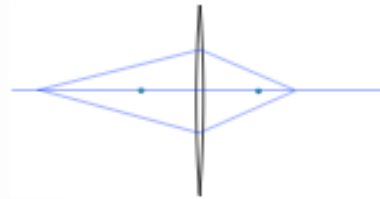


(b)

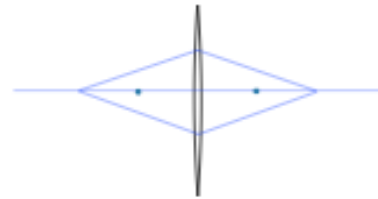
A convex tin lens



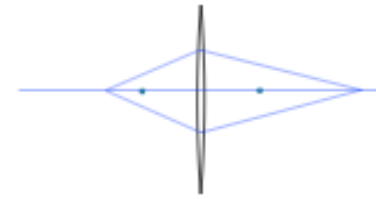
(a)



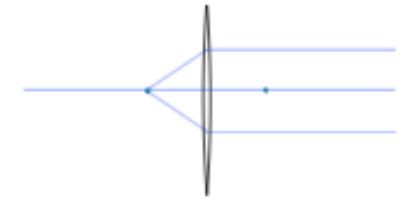
(b)



(c)



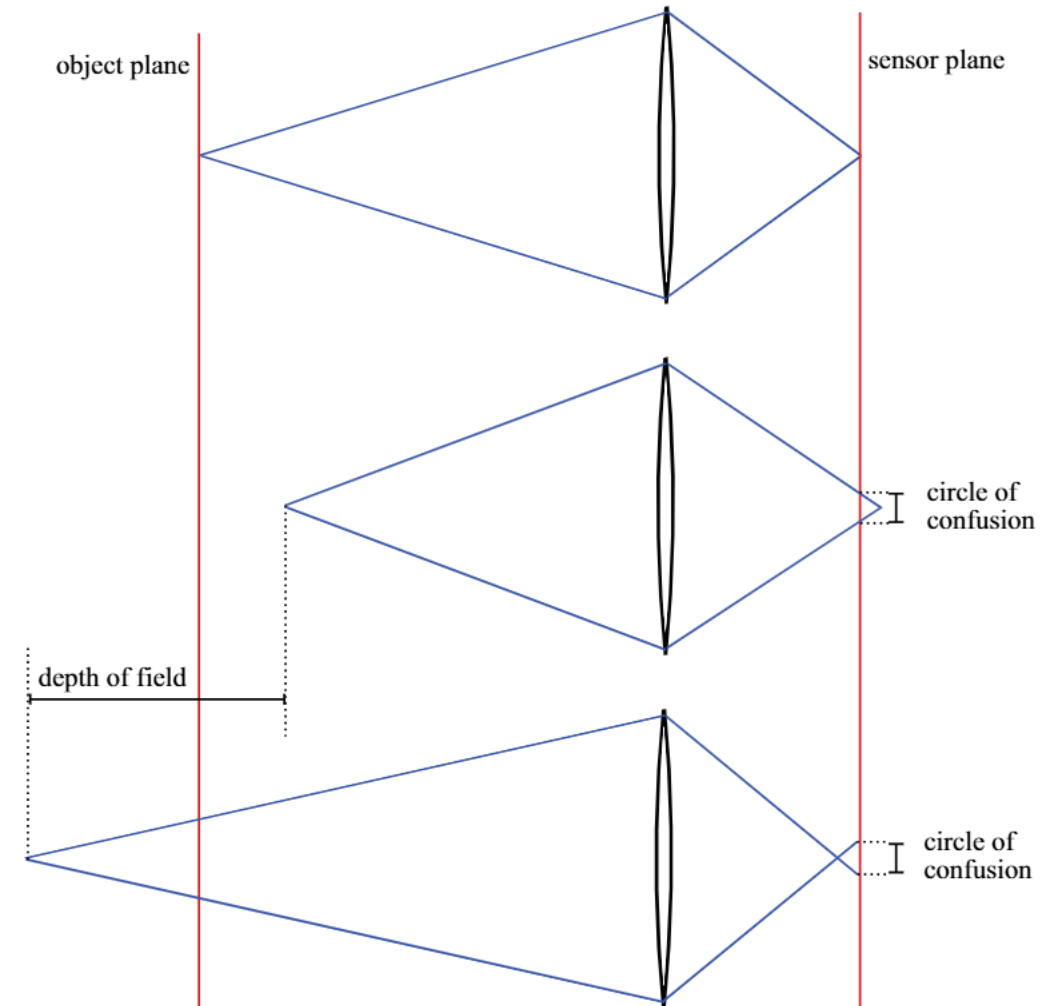
(d)



(e)

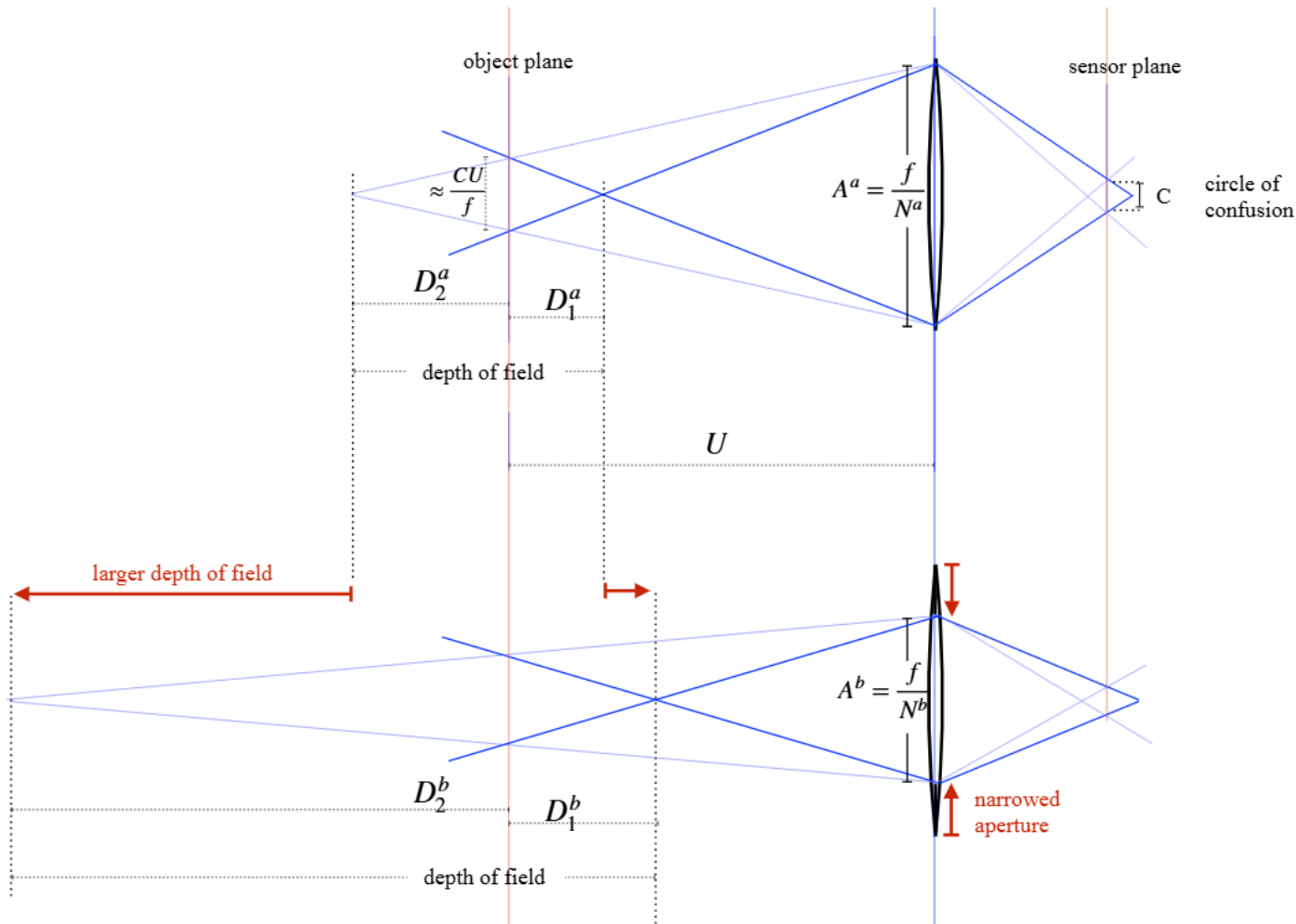
Depth of field

...in infinity, the area where the
distance for that aperture, the
scales on a lens barrel
perforal distance opposi
u are using. If you the
the depth of field wil
ce to infinity. For
amera has a hyperf
e focus at 18 feet,



- Top image CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=330435>
- Bottom image By NightWolf1223 - Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=118009181>

Depth of field



F-number (N) = 2

F-number (N) = 4

F-number (N) = 8



(a) f/2.0



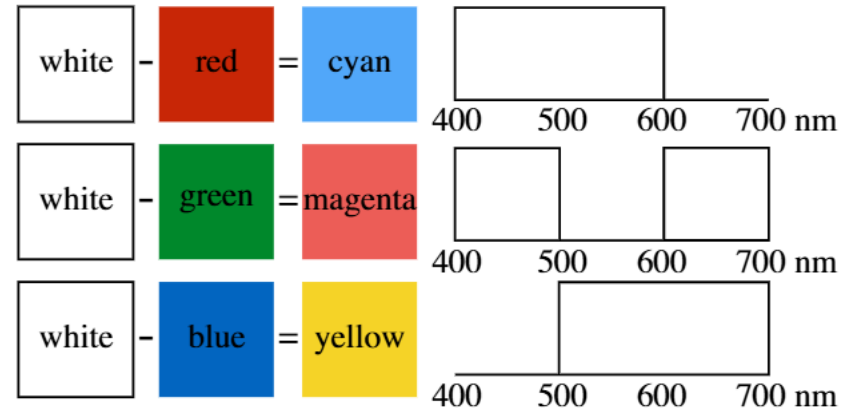
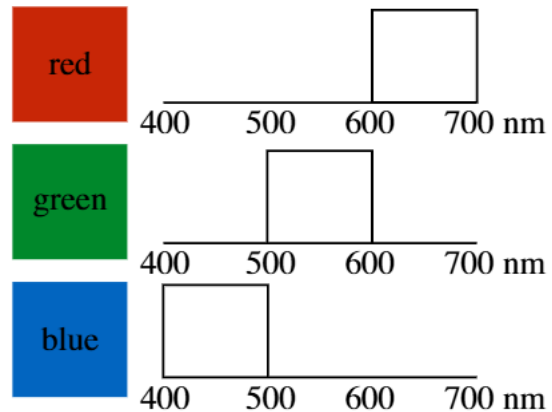
(b) f/4.0



(c) f/8.0



Color

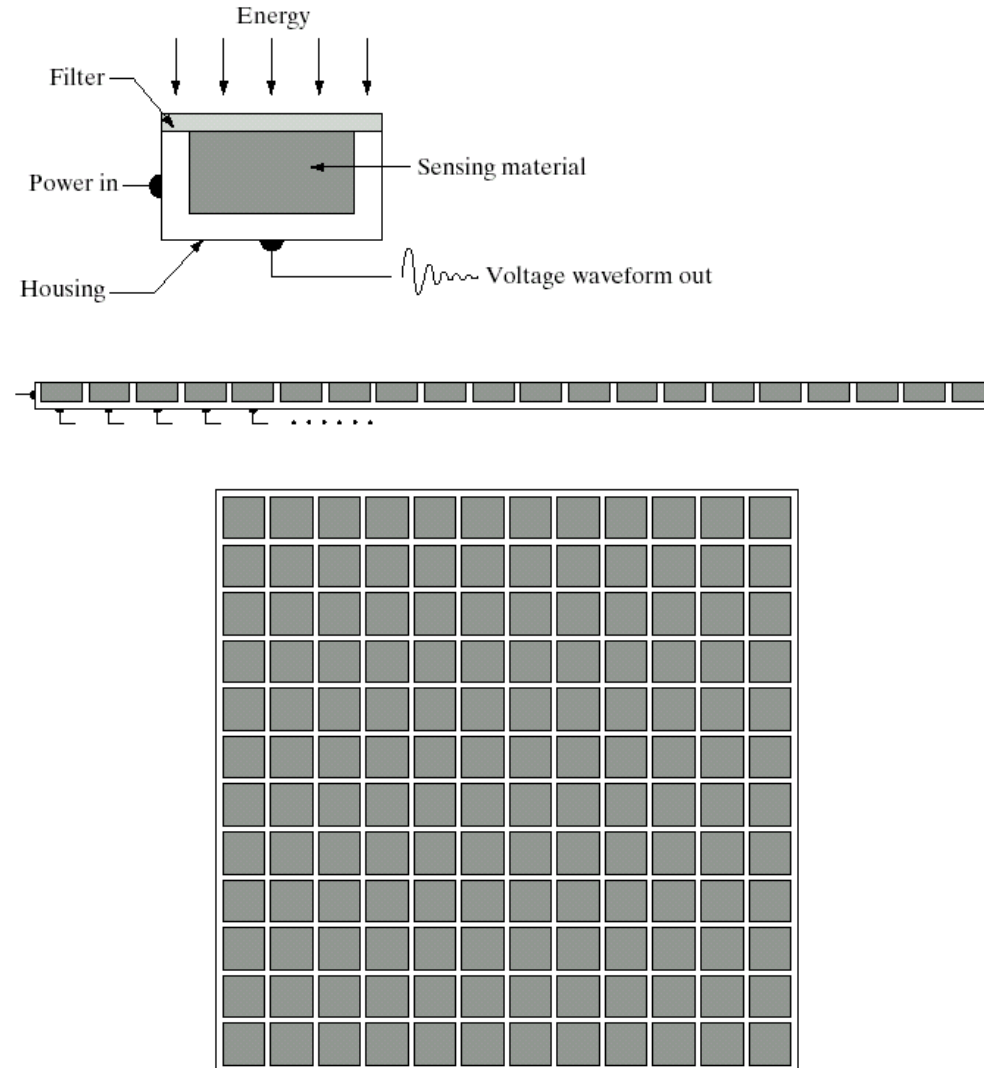


Camera Sensors

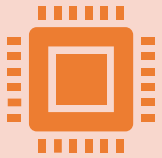
a
b
c

FIGURE 2.12

(a) Single imaging sensor.
(b) Line sensor.
(c) Array sensor.



Fundamentals of Camera Sensors



Modern cameras use solid-state electronic devices containing millions of photodetector sites (pixels)



Two main categories dominate machine vision:

Charge-Coupled Device (CCD)
Complementary Metal Oxide
Semiconductor (CMOS)

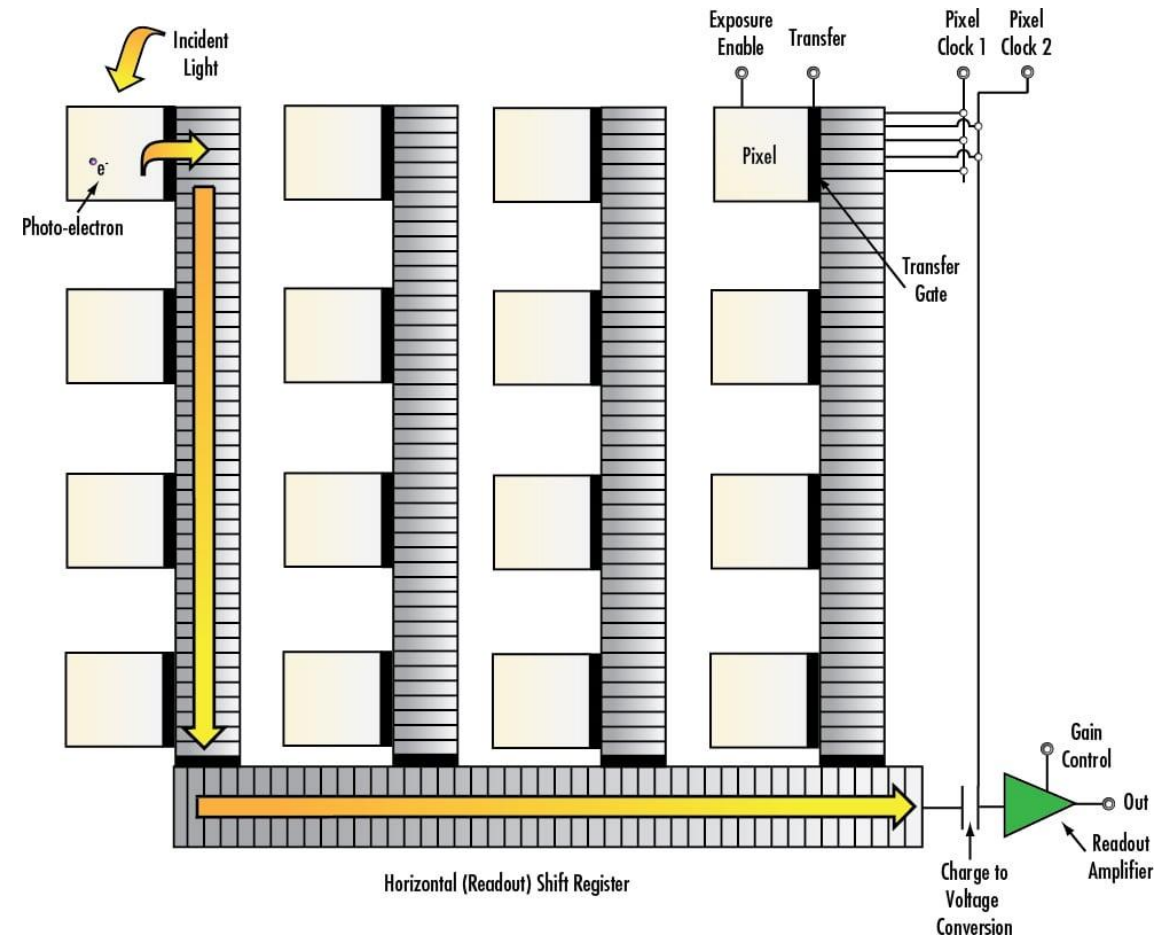
Camera Sensors

CCD Sensors: Key Characteristics

charge-coupled device (CCD)

Invented in 1969 at Bell Labs

- Operating Principle:
 - Uses "bucket brigade" method to move charge packets
 - Charge packets converted to voltage by capacitor
 - Analog device with digital conversion via ADC
- Advantages:
 - High sensitivity
 - Excellent pixel-to-pixel consistency
 - High uniformity
 - Superior image quality
- Limitations:
 - Speed constraints due to charge transfer
 - Higher power consumption

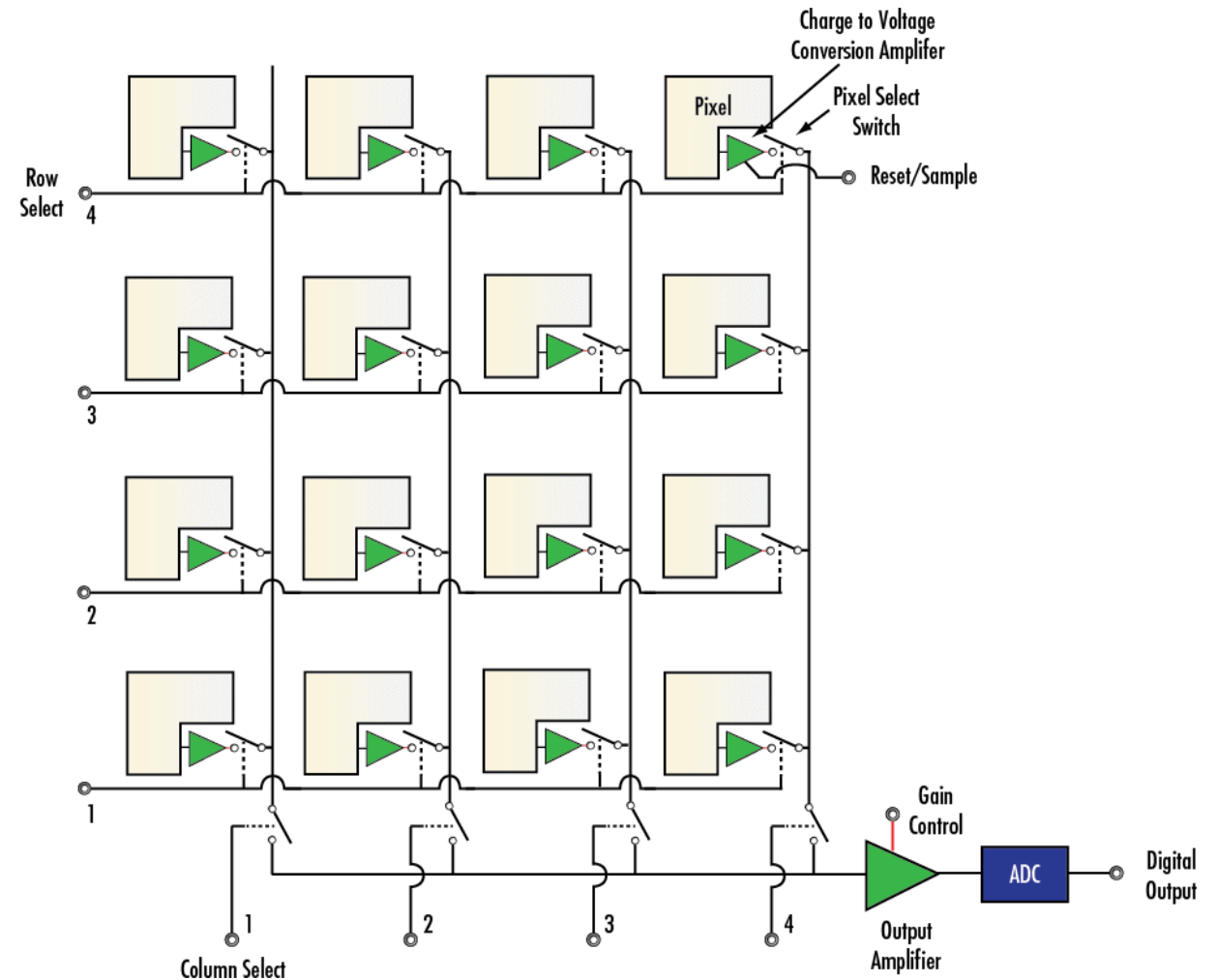


Camera Sensors

CMOS Sensors: Key Characteristics

Invented in 1963, widely adopted in 1990s

- Operating Principle:
 - Charge converted to voltage at pixel site
 - Each pixel contains photodiode and three transistors
 - Inherently digital device
- Advantages:
 - Higher speed capability
 - Lower power consumption
 - Better handling of high light levels
 - Compact size
- Limitations:
 - Lower sensitivity
 - Higher fixed-pattern noise
 - Lower signal-to-noise ratio
 - Less uniform image quality



Frame Rate and Shutter Types

Frame Rate Factors:

- Sensor readout speed
- Interface data transfer rate
- Number of pixels
- Resolution settings

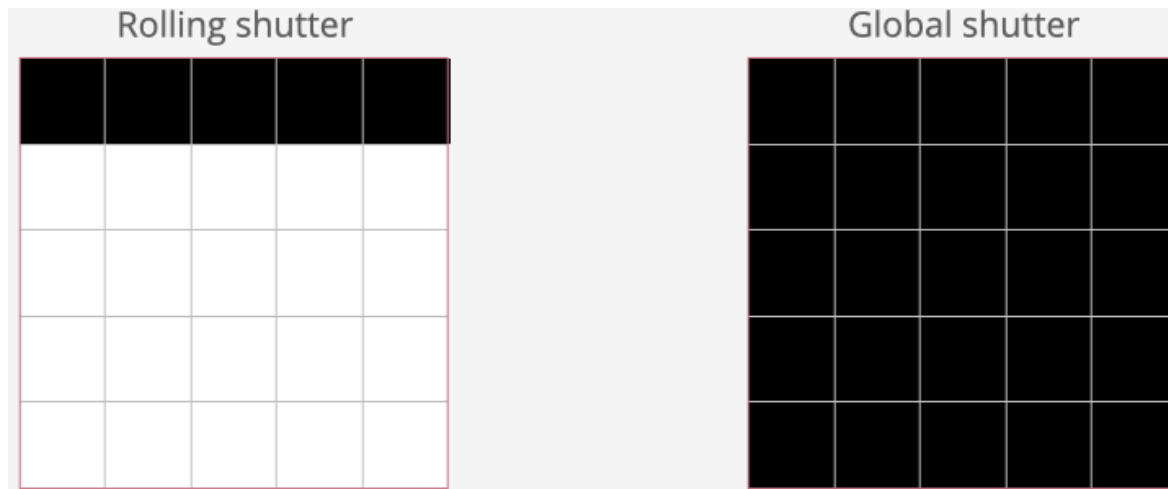
Shutter Types:

1. Global Shutter:

- All pixels exposed simultaneously
- Better for moving objects
- Available in both CCD and modern CMOS

2. Rolling Shutter:

- Sequential exposure and readout
- Can cause distortion in moving objects
- Common in basic CMOS sensors



Sensor Sizes and Standards

- Common formats: 1/4", 1/3", 1/2", 2/3", 1", 1.2"
 - Larger sensors = greater field of view
 - Must match lens capabilities
- Standard 4:3 aspect ratio

Digital image acquisition processes

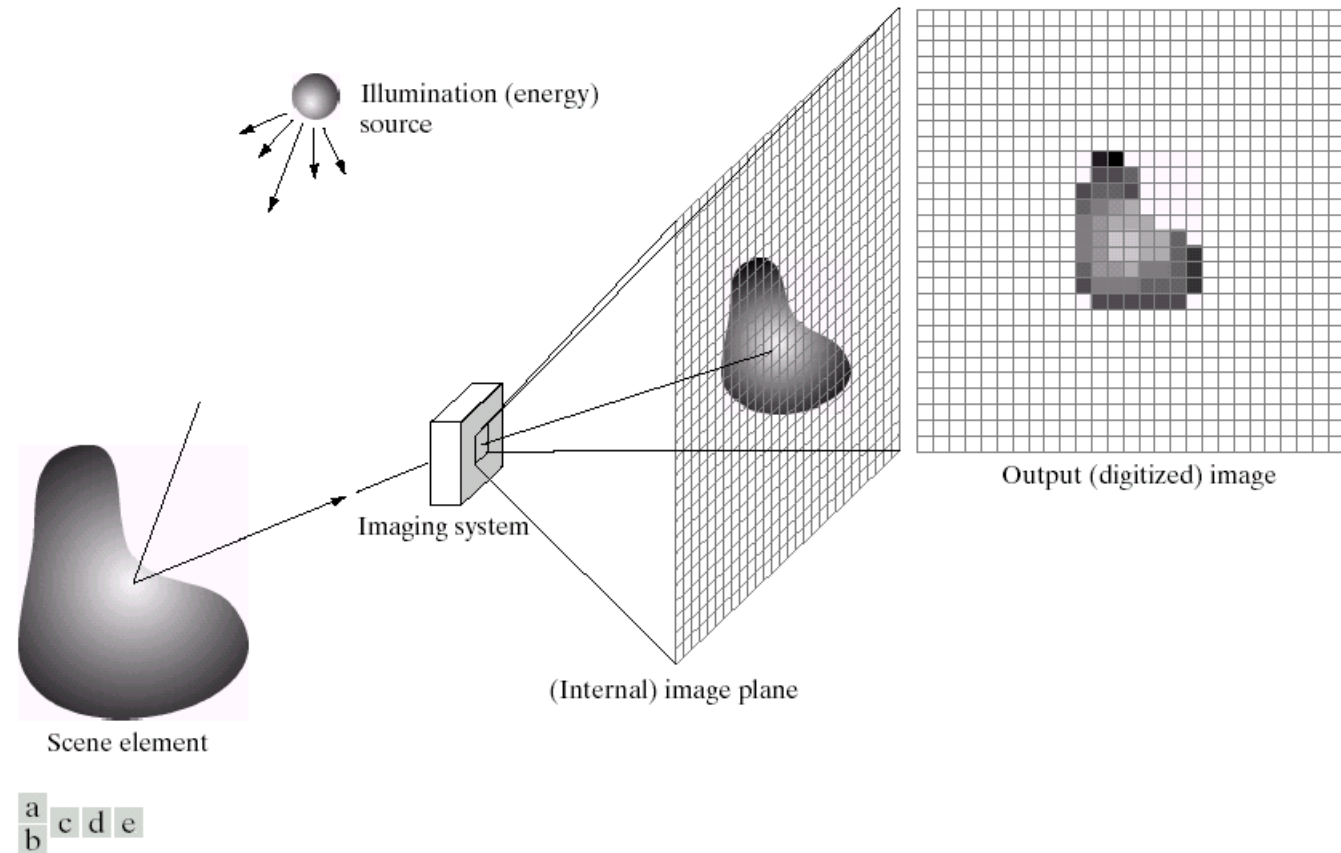


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy (“illumination”) source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Sensor Features: Understanding Pixels

- Pixels are discrete photodetector sites
- Key Characteristics:
 - Typically square in digital cameras
 - Size range: 3-10 μ m
 - Quantum efficiency: 30-60%
 - Larger pixels = higher charge capacity and better SNR
 - Smaller pixels = higher resolution but more noise